

Expanded Disease Knowledge Base

RAG-ready structured dataset • 35 diseases/conditions • Explicit causes and precautions

Important: This is educational information for a retrieval system, not a diagnostic or treatment tool. Statistics are tied to the year stated and may change as new estimates are published.

Fields in every record: Disease name • Type • Cause • Transmission/spread • Symptoms • Precautions • Key data

1. Tuberculosis (TB)

Disease	Tuberculosis (TB)
Type	Infectious • Bacterial
CAUSE	Mycobacterium tuberculosis bacteria.
Transmission / Spread	Spread mainly through the air when a person with infectious pulmonary TB coughs, sneezes or speaks.
Symptoms	Persistent cough, fever, night sweats, weight loss, weakness, chest pain and sometimes coughing blood.
PRECAUTIONS	Avoid prolonged close exposure to untreated infectious TB, improve ventilation, seek testing for symptoms/exposure, follow prescribed treatment, and use masks/respiratory precautions when advised.
Key data	WHO estimates 10.7 million people developed TB and 1.23 million died in 2024.

2. Malaria

Disease	Malaria
Type	Infectious • Parasitic
CAUSE	Plasmodium parasites.
Transmission / Spread	Usually transmitted by bites of infected female Anopheles mosquitoes.
Symptoms	Fever, chills and headache; severe disease can cause confusion, seizures, breathing problems, jaundice or organ failure.
PRECAUTIONS	Use insecticide-treated bed nets, repellents and protective clothing; reduce mosquito breeding sites; use preventive medicines when medically indicated.
Key data	WHO estimated 282 million cases and 610,000 deaths globally in 2024.

3. Dengue

Disease	Dengue
Type	Infectious • Viral
CAUSE	Dengue virus (DENV).
Transmission / Spread	Usually transmitted by infected Aedes mosquitoes.
Symptoms	High fever, headache, pain behind the eyes, muscle/joint pain, nausea, vomiting and rash; severe dengue can cause bleeding, shock or organ problems.
PRECAUTIONS	Prevent mosquito bites, remove standing water, use screens/repellents and seek urgent care for warning signs such as severe abdominal pain or persistent vomiting.
Key data	WHO reported 14.6 million dengue cases in 2024 and estimates 100–400 million infections occur each year.

4. Rabies

Disease	Rabies
Type	Infectious • Viral • Zoonotic
CAUSE	Rabies virus.

Transmission / Spread	Usually transmitted through saliva from an infected animal, commonly via bites or scratches.
Symptoms	Fever and abnormal sensations near the wound may progress to agitation, difficulty swallowing, hydrophobia, paralysis and coma.
PRECAUTIONS	Avoid contact with unfamiliar animals; vaccinate dogs where recommended; immediately wash bites/scratches with soap and water and obtain urgent post-exposure medical care.
Key data	Rabies is virtually 100% fatal after clinical symptoms appear, but prompt post-exposure prophylaxis can prevent disease.

5. Measles

Disease	Measles
Type	Infectious • Viral
CAUSE	Measles virus.
Transmission / Spread	Highly contagious airborne/respiratory transmission.
Symptoms	High fever, cough, runny nose, red eyes, Koplik spots and a widespread rash.
PRECAUTIONS	Routine measles vaccination is the main prevention; avoid exposure during outbreaks and follow public-health guidance.
Key data	WHO estimated about 95,000 measles deaths globally in 2024.

6. HIV Infection / AIDS

Disease	HIV Infection / AIDS
Type	Infectious • Viral
CAUSE	Human immunodeficiency virus (HIV).
Transmission / Spread	Transmission through infected blood, semen, vaginal fluids or breast milk, including sexual and mother-to-child transmission.
Symptoms	Early infection may cause flu-like symptoms or none; untreated disease can cause weight loss, fever, diarrhoea, cough and opportunistic infections.
PRECAUTIONS	Use condoms/barrier protection, test regularly when at risk, use PrEP when appropriate, never share injection equipment, and use medical prevention for mother-to-child transmission.
Key data	WHO estimated 41.0 million people were living with HIV at the end of 2025.

7. Hepatitis B

Disease	Hepatitis B
Type	Infectious • Viral
CAUSE	Hepatitis B virus (HBV).
Transmission / Spread	Transmission can occur at birth, through infected blood/body fluids, sexual contact and unsafe injections.
Symptoms	Often no symptoms; fatigue, nausea, abdominal discomfort, dark urine and jaundice can occur. Chronic infection can lead to cirrhosis and liver cancer.
PRECAUTIONS	Get hepatitis B vaccination, use sterile medical/injection equipment, screen blood, use barrier protection where appropriate and ensure infected pregnant people receive recommended care.
Key data	WHO estimated 240 million people had chronic hepatitis B infection in 2024.

8. Hepatitis C

Disease	Hepatitis C
Type	Infectious • Viral
CAUSE	Hepatitis C virus (HCV).
Transmission / Spread	Primarily transmitted through contact with infected blood.
Symptoms	Often asymptomatic; possible fatigue, nausea, abdominal pain, dark urine and jaundice.
PRECAUTIONS	Use sterile injection and medical equipment, ensure blood is screened, and avoid sharing needles or other items contaminated with blood. No effective vaccine is currently available.
Key data	Direct-acting antivirals can cure more than 95% of people with hepatitis C.

9. Cholera

Disease	Cholera
Type	Infectious • Bacterial
CAUSE	Toxigenic <i>Vibrio cholerae</i> bacteria.

Transmission / Spread	Usually acquired by drinking water or eating food contaminated with feces containing the bacteria.
Symptoms	Sudden watery diarrhoea and vomiting; severe dehydration can develop rapidly.
PRECAUTIONS	Use safe drinking water, sanitation and hand hygiene; cook food safely; follow outbreak vaccination and public-health advice where applicable.
Key data	Severe cholera can become fatal within hours without rehydration treatment.

10. Typhoid Fever

Disease	Typhoid Fever
Type	Infectious • Bacterial
CAUSE	Salmonella Typhi bacteria.
Transmission / Spread	Usually spread through food or water contaminated with feces from an infected person.
Symptoms	Prolonged fever, headache, fatigue, abdominal pain, diarrhoea or constipation and sometimes rash.
PRECAUTIONS	Use safe water and food, wash hands, maintain sanitation, and receive typhoid vaccination when recommended.
Key data	Antibiotic resistance is an important challenge in typhoid treatment.

11. Influenza (Flu)

Disease	Influenza (Flu)
Type	Infectious • Viral
CAUSE	Influenza viruses, mainly types A and B in seasonal human disease.
Transmission / Spread	Respiratory droplets and aerosols from infected people; contaminated hands/surfaces can also contribute.
Symptoms	Sudden fever, cough, sore throat, body aches, headache and fatigue; some people develop complications such as pneumonia.
PRECAUTIONS	Annual vaccination where recommended, hand hygiene, good ventilation, staying home when sick and avoiding close contact with vulnerable people during illness.
Key data	Seasonal influenza causes substantial illness worldwide each year; burden varies by season and population.

12. COVID-19

Disease	COVID-19
Type	Infectious • Viral
CAUSE	SARS-CoV-2.
Transmission / Spread	Primarily through respiratory particles released by infected people, especially during close or prolonged indoor contact.
Symptoms	Fever, cough, sore throat, fatigue, headache, loss/change of smell or taste and breathing difficulty; symptoms vary.
PRECAUTIONS	Stay up to date with recommended vaccination, improve ventilation, stay home when ill and use appropriate masks in higher-risk settings.
Key data	COVID-19 can range from asymptomatic infection to severe disease and death; risk is higher in older and medically vulnerable people.

13. Pneumonia

Disease	Pneumonia
Type	Infectious/Noninfectious • Respiratory
CAUSE	Many causes including bacteria, viruses and fungi; Streptococcus pneumoniae is a common bacterial cause.
Transmission / Spread	Depends on cause; respiratory infections can spread through droplets/aerosols or contact.
Symptoms	Cough, fever, chills, chest pain, rapid or difficult breathing and fatigue; older adults may have less typical symptoms.
PRECAUTIONS	Vaccination where recommended, hand hygiene, good ventilation, avoiding tobacco smoke and managing chronic conditions reduce risk.
Key data	Pneumonia can be caused by many pathogens, so prevention and treatment depend on the cause.

14. Meningitis

Disease	Meningitis
Type	Infectious • Neurological

CAUSE	Inflammation of the membranes around the brain and spinal cord; causes include bacteria, viruses, fungi and parasites.
Transmission / Spread	Depends on cause; some bacterial meningitis spreads through respiratory/oral secretions during close contact.
Symptoms	Fever, severe headache, neck stiffness, vomiting, confusion, sensitivity to light and sometimes rash.
PRECAUTIONS	Vaccination against relevant meningococcal, pneumococcal and Hib infections; reduce close exposure during outbreaks and seek urgent care for symptoms.
Key data	Bacterial meningitis is a medical emergency requiring rapid treatment.

15. Polio

Disease	Polio
Type	Infectious • Viral
CAUSE	Poliovirus.
Transmission / Spread	Mainly fecal-oral transmission; less commonly through contaminated food/water.
Symptoms	Many infections cause no symptoms; severe disease can cause meningitis or irreversible paralysis.
PRECAUTIONS	Vaccination is the key prevention; maintain sanitation and hygiene and follow outbreak vaccination campaigns.
Key data	Polio vaccination has reduced global disease dramatically, but eradication requires continued vaccination and surveillance.

16. Chickenpox

Disease	Chickenpox
Type	Infectious • Viral
CAUSE	Varicella-zoster virus.
Transmission / Spread	Highly contagious through respiratory particles and direct contact with blister fluid.
Symptoms	Itchy blister-like rash, fever, tiredness and headache.
PRECAUTIONS	Varicella vaccination where recommended; avoid close contact with susceptible high-risk people while infectious.
Key data	Most healthy children recover, but complications can occur, especially in adults, pregnant people and immunocompromised people.

17. Mumps

Disease	Mumps
Type	Infectious • Viral
CAUSE	Mumps virus.
Transmission / Spread	Respiratory droplets and saliva from infected people.
Symptoms	Fever, headache, muscle aches and painful swelling of the salivary glands, especially near the jaw.
PRECAUTIONS	MMR vaccination is the main prevention; avoid close contact during infectious periods.
Key data	Complications can include meningitis, encephalitis and hearing problems.

18. Whooping Cough (Pertussis)

Disease	Whooping Cough (Pertussis)
Type	Infectious • Bacterial
CAUSE	Bordetella pertussis bacteria.
Transmission / Spread	Respiratory droplets from coughing or sneezing.
Symptoms	Initially cold-like symptoms followed by severe coughing fits, sometimes with a characteristic whoop or vomiting.
PRECAUTIONS	Vaccination is the main prevention; early diagnosis and treatment can reduce spread; protect infants through recommended maternal vaccination.
Key data	Young infants are at greatest risk of severe complications.

19. Leprosy

Disease	Leprosy
Type	Infectious • Bacterial
CAUSE	Mycobacterium leprae and, in some regions, M. lepromatosis.
Transmission / Spread	Likely spread through prolonged, close contact involving respiratory droplets; transmission is not highly efficient.
Symptoms	Skin patches with reduced sensation, numbness, nerve thickening, weakness and sometimes eye problems.

PRECAUTIONS	Early diagnosis and complete multidrug therapy reduce transmission and complications; avoid prolonged close contact with untreated cases.
Key data	Leprosy is curable with multidrug therapy.

20. Yellow Fever

Disease	Yellow Fever
Type	Infectious • Viral
CAUSE	Yellow fever virus.
Transmission / Spread	Bites of infected mosquitoes.
Symptoms	Fever, headache, muscle aches, nausea and vomiting; severe disease can cause jaundice, bleeding and organ failure.
PRECAUTIONS	Vaccination for people in risk areas or travelers when indicated, plus mosquito-bite prevention.
Key data	Yellow fever vaccination provides long-lasting protection for most people.

21. Japanese Encephalitis

Disease	Japanese Encephalitis
Type	Infectious • Viral
CAUSE	Japanese encephalitis virus.
Transmission / Spread	Usually transmitted by infected Culex mosquitoes; pigs and wading birds are important hosts in the transmission cycle.
Symptoms	Most infections cause no symptoms; encephalitis can cause fever, headache, confusion, seizures and neurological problems.
PRECAUTIONS	Vaccination where recommended, mosquito-bite prevention and reducing exposure to mosquitoes.
Key data	Japanese encephalitis is a major cause of viral encephalitis in parts of Asia and the Western Pacific.

22. Chikungunya

Disease	Chikungunya
Type	Infectious • Viral
CAUSE	Chikungunya virus.
Transmission / Spread	Primarily transmitted by infected Aedes mosquitoes.
Symptoms	Sudden fever and severe joint pain, often with rash, headache or muscle pain.
PRECAUTIONS	Prevent mosquito bites and remove standing water; use repellents, protective clothing and screens.
Key data	Joint pain can persist for weeks or months in some patients.

23. Zika Virus Disease

Disease	Zika Virus Disease
Type	Infectious • Viral
CAUSE	Zika virus.
Transmission / Spread	Mainly through infected Aedes mosquitoes; sexual and mother-to-child transmission can also occur.
Symptoms	Often mild or asymptomatic; possible rash, fever, conjunctivitis, joint pain and headache.
PRECAUTIONS	Prevent mosquito bites and follow sexual and pregnancy-related public-health guidance in areas with transmission.
Key data	Zika infection during pregnancy can cause congenital abnormalities.

24. Asthma

Disease	Asthma
Type	Noncommunicable • Chronic respiratory
CAUSE	Chronic airway inflammation and variable narrowing; triggers differ between people.
Transmission / Spread	Not contagious.
Symptoms	Wheezing, cough, shortness of breath and chest tightness that can vary over time.

PRECAUTIONS	Avoid personal triggers such as tobacco smoke and relevant allergens, reduce air-pollution exposure and use prescribed controller medicines correctly.
Key data	WHO estimated 363 million people had asthma in 2023.

25. COPD

Disease	COPD
Type	Noncommunicable • Chronic respiratory
CAUSE	Persistent airflow limitation, commonly linked to tobacco smoke and air pollution.
Transmission / Spread	Not contagious.
Symptoms	Long-term cough, phlegm, breathlessness, wheezing and flare-ups.
PRECAUTIONS	Do not smoke, avoid second-hand smoke and harmful air pollution, reduce occupational exposures and follow vaccination advice.
Key data	WHO reports 3.4 million COPD deaths in 2023.

26. Type 2 Diabetes

Disease	Type 2 Diabetes
Type	Noncommunicable • Metabolic
CAUSE	Insulin resistance combined with inadequate insulin production.
Transmission / Spread	Not contagious.
Symptoms	Often develops gradually without symptoms; increased thirst, frequent urination, fatigue and blurred vision can occur.
PRECAUTIONS	Regular physical activity, healthy eating, healthy body weight, avoiding tobacco and routine screening when at risk.
Key data	WHO reports that about 830 million people were living with diabetes in 2022.

27. Hypertension

Disease	Hypertension
Type	Noncommunicable • Cardiovascular
CAUSE	Persistently elevated blood pressure; often caused by multiple interacting factors.
Transmission / Spread	Not contagious.
Symptoms	Usually no symptoms; very high blood pressure can cause severe headache, chest pain, breathing problems or neurological symptoms.
PRECAUTIONS	Measure blood pressure, reduce excess salt, exercise, maintain healthy weight, avoid tobacco and limit alcohol.
Key data	WHO estimates 1.4 billion adults had hypertension in 2024.

28. Stroke

Disease	Stroke
Type	Noncommunicable • Neurological emergency
CAUSE	Blocked blood flow to the brain or bleeding in the brain.
Transmission / Spread	Not contagious.
Symptoms	Sudden facial droop, arm weakness, speech difficulty, vision problems, dizziness or loss of coordination.
PRECAUTIONS	Control blood pressure, cholesterol and diabetes; avoid tobacco; exercise; maintain a healthy diet; manage atrial fibrillation when present.
Key data	WHO estimated 11.9 million new strokes in 2021.

29. Heart Attack

Disease	Heart Attack
Type	Noncommunicable • Cardiovascular emergency
CAUSE	Usually a blocked coronary artery caused by a blood clot forming on a ruptured atherosclerotic plaque.
Transmission / Spread	Not contagious.

Symptoms	Chest pressure/pain, shortness of breath, sweating, nausea or discomfort spreading to the arm, jaw, neck or back.
PRECAUTIONS	Do not smoke, control blood pressure/cholesterol/diabetes, exercise regularly, maintain a healthy diet and follow prescribed preventive medicines.
Key data	WHO reports ischaemic heart disease caused about 9 million deaths in 2021.

30. Cancer

Disease	Cancer
Type	Noncommunicable • Malignant disease
CAUSE	Abnormal cells grow uncontrollably; causes vary by cancer type.
Transmission / Spread	Cancer itself is generally not contagious.
Symptoms	Depends on cancer type; possible unexplained weight loss, persistent lump, bleeding, prolonged cough, bowel changes or persistent pain.
PRECAUTIONS	Avoid tobacco, limit alcohol, maintain healthy weight and activity, protect skin from UV radiation, and follow recommended HPV/hepatitis B vaccination and screening.
Key data	WHO reports nearly 10 million cancer deaths worldwide in 2024.

31. Chronic Kidney Disease (CKD)

Disease	Chronic Kidney Disease (CKD)
Type	Noncommunicable • Kidney
CAUSE	Long-term kidney damage or reduced kidney function from conditions such as diabetes, hypertension and other diseases.
Transmission / Spread	Not contagious.
Symptoms	Often no symptoms early; later may cause swelling, fatigue, changes in urination, nausea or itching.
PRECAUTIONS	Control blood pressure and diabetes, avoid unnecessary kidney-harming medicines, avoid tobacco and maintain a healthy lifestyle.
Key data	CKD can progress to kidney failure and cardiovascular complications; early detection can slow progression.

32. Osteoporosis

Disease	Osteoporosis
Type	Noncommunicable • Musculoskeletal
CAUSE	Loss of bone strength and density, influenced by aging, hormones, nutrition, medicines and other conditions.
Transmission / Spread	Not contagious.
Symptoms	Often silent until a fracture occurs; fractures commonly affect the hip, spine or wrist.
PRECAUTIONS	Adequate calcium/vitamin D, weight-bearing exercise, fall prevention, avoiding tobacco and limiting alcohol; assess high-risk people medically.
Key data	Osteoporosis increases fracture risk, particularly with aging.

33. Alzheimer Disease

Disease	Alzheimer Disease
Type	Noncommunicable • Neurological
CAUSE	A progressive brain disorder involving abnormal protein changes and loss of nerve-cell function; exact causes are complex.
Transmission / Spread	Not contagious.
Symptoms	Progressive memory loss, difficulty with thinking or language, impaired daily functioning and later behavioral or physical changes.
PRECAUTIONS	Manage cardiovascular risk factors, exercise, avoid tobacco, maintain social/cognitive activity and address hearing/vision problems; no guaranteed prevention exists.
Key data	Dementia is a major cause of disability and dependency in older adults; Alzheimer disease is its most common cause.

Suggested RAG Document Schema

name: disease name

type: disease category

cause: primary cause/agent

transmission: how it spreads, or 'not contagious'

symptoms: common symptoms

precautions: prevention/risk-reduction actions

data: dated epidemiological fact

Suggested chunking: Keep each disease as one document for broad retrieval. For question-answering over specific fields, also store Cause and Precautions as searchable metadata or separate child chunks.

Source base: World Health Organization (WHO) fact sheets and health-topic resources. For a production medical RAG system, refresh sources regularly and add clinical guidelines relevant to your target country.